

Samiran Kawtikwar

✉ samkawtikwar@gmail.com | +1 217 200 1346 | , ,

Interests: Discrete Optimization, Parallel algorithms, Mathematical Modeling, GPU acceleration

EDUCATION

University of Illinois Urbana-Champaign
PhD Operations Research
2020 - August 2025
MS Industrial Engg. 2020 - 2022

Indian Institute of Technology Delhi
B. Tech Production and Industrial Engineering
2014 - 2018 | Delhi, India
Department Rank 3

COURSEWORK

Integer Programming
Applied Parallel Programming
Convex Optimization
Combinatorial Optimization
Large scale Network Opt
Optimization Under Uncertainty
Machine Learning

Teaching Assistant
Facilities Planning and Design
Big Graphs and Social Networks
Production Planning & Control
Intro to Operations Research

FOCUS AREAS

High-Performance Computing
Large-Scale Optimization
Parallel Programming
Discrete Optimization (MILP)

SKILLS

C++, CUDA, OpenMP, Python, CuPy, Java, MATLAB, R, Git, Gurobi, Cvxpy, AMPL, SQL

POSITIONS

at UIUC
Graduate Mentor, UIUC
at Schlumberger
Empowerment team leader, SLB
at IIT Delhi
Undergraduate Secretary, AIC
Publicity coordinator, Literati

INTERESTS

Badminton, Aquatics, Soccer, Traveling

PROFESSIONAL EXPERIENCE

Wayfair | Senior Operations Research Scientist
Transportation and Delivery (Boston, USA) | Aug 25 - Present
- Developed an integrated mathematical optimization model to jointly optimize weekly trailer allocation and labor scheduling across Wayfair's fulfillment operations.
- Collaborated with Engineering and Data Science teams to productionize the optimization engine, integrate model-driven planning decisions into Wayfair's logistics operations.
- Translated non-technical business requirements into optimization objectives and constraints, evaluate operational trade-offs and explain planning decisions.

ExxonMobil | Mathematical Optimization and Data Science Intern
Supply Chain and Market Modelling (Houston, USA) | May - Aug 24
- Enhanced Long Term Portfolio Model via reformulation and preprocessing, at same quality.
- Built a CPU parallelization framework to simultaneously evaluate multiple pricing forecasts.

Schlumberger | Field Operations Engineer
Subsea Landing String Services (India, China, France) | July 18 – July 20
- Delivered efficient offshore production operations for global clients on Asia's deepest wells.
- Received highest ratings from clients for efficient performance, safety, and communication.

RELEVANT PROJECTS

Large-Scale Discrete Optimization
GPU accelerated Branch and Bound through Best First Search
- Developed a single kernel branch and bound framework that performs Best first search.
- Solved Resource Constrained Assignment Problem using the framework ($\sim 446\times$ speedup).

GPU Accelerated large Linear Assignment Problem Solver
- HyLAC is the fastest linear assignment problem solver, part of the NVIDIA RAPIDS library.
- Geomean time speedup of **6.14x** over the previous state-of-the-art implementations.

GPU Accelerated crossover algorithm
- First implementation of a high performance GPU-accelerated crossover algorithm.
- Parallel algorithm with $100\times$ fewer primal/dual pushes than simplex based algorithms.

Big Graphs Analysis
Accelerated Subgraph enumeration in CUDA
- Developed the fastest GPU Subgraph Enumerator using CUDA with **7.8x** improvement.
- Deployed the application on a supercomputer cloud cluster and achieved strong scaling.

PUBLICATIONS

1. **IEEE Transactions on Parallel and Distributed Computing (TPDS' 24)**: GPU-Accelerated Branch and Bound through Best First Search. (Under review)
2. **Journal of Parallel and Distributed Computing (JPDC' 23)**: HyLAC - Hybrid Linear Assignment solver in CUDA.
3. **International Conference on Parallel Processing (ICPP '23)**: BEEP - Balanced Efficient subgraph Enumeration in Parallel.
4. **International Conference on Information Fusion (FUSION' 23)**: Multi-Target Tracking with GPU-Accelerated Data Association Engine.

AWARDS AND ACHIEVEMENTS

2022, 23, 24	Awarded **excellent teaching assistant** at UIUC for 4 consecutive semesters.
2020-2021	Recipient of **Phil Ryan Graduate Fellowship** for excellent academic record.
2017-2018	**Semester Merit Award** for outstanding academic performance at IIT Delhi.
2014-2018	Awarded **MCM scholarship** with tuition waiver for all 4 undergraduate years.